Research Statement

My primary mathematical interests lie in algebraic geometry, with a specific focus on the positivity of linear series and the geometry of toric and tropical varieties. I am fascinated by the way these areas use combinatorics to connect disparate mathematical ideas. My long-term goal is to contribute original research to the field and to mentor students who share my passion for mathematics.

My initial exposure to algebraic geometry came during my undergraduate thesis at Sun Yat-sen University. This project introduced me to toric varieties, and I was captivated by how they transform complicated geometric structures into accessible combinatorial data derived from polytopes and fans—serving as a kind of dictionary translating between algebraic geometry and combinatorics. My work led to the paper "On covering simplices by dilations in dimensions 3 and 4" on arXiv. Through this process, I began learning the fundamentals of algebraic geometry and discovered great joy in delving deeply into a problem until a moment of clarity emerges. I often describe this process as "peeling an onion," where each layer reveals a new level of understanding. This research experience sparked my enduring interest in algebraic geometry, especially its combinatorial aspects.

Because of my outstanding academic record and overall performance, I was recommended for admission to the master's program without taking the entrance examination. During my graduate studies, I completed advanced courses such as Riemannian Geometry, Algebraic Topology, and Commutative Algebra with excellent results, which strengthened my foundation in pure mathematics. Furthermore, I gave a series of talks in small reading seminars on Introduction to Toric Varieties and Graded Syzygies at my university. These experiences have allowed me to resolve my questions and gain clearer, deeper insights through discussions with others. Serving as a speaker also enhanced my communication skills—teaching me to focus on the core of a problem and articulate complex ideas with clarity. For my strong academic performance, I received the Graduate Scholarship from my university for three consecutive years.

Beyond coursework, I have also actively participated in various academic activities to broaden my exposure to the wider mathematical world and to appreciate its different flavors. For instance, I learned about multiplier ideals, V-filtrations, and K-stability theory in singularities during the Algebraic Geometry Summer School held in Shanghai in 2024, as well as stable reduction techniques in a mini-course taught by Prof. Qing Liu.

My current research focuses on toric geometry, particularly the positivity of linear series and syzygies. The concept of syzygies originated in the 19th century with the work of Sylvester, Cayley, and Hilbert, describing the shape of free resolutions and Betti numbers in commutative algebra. From the 20th century onward, syzygies became a bridge between algebraic geometry and commutative algebra, connecting the structure of minimal free resolutions to the geometry of embedded projective varieties. In particular, syzygies capture subtle information about how a variety is embedded—its projective normality, generation by quadrics, and higher-order relations. My research has exposed me to a variety of topics, including GIT theory, stability of syzygy bundles, Klyachko's classification, and resolutions of toric subvarieties. This work has taught me to navigate both abstract theory and concrete computation, and to approach problems from multiple perspectives. More importantly, it has shown me the essence of research: to remain curious amid uncertainty and to find meaning in the gradual discovery. I have come to truly enjoy walking through this "forest" of mathematics—experimenting, learning through trial and error, and uncovering new structures along the way.

In addition, during the first year of my master's program, I explored tropical geometry and wrote a survey on "Curve Counting and Tropical Geometry" for a graduate-level course. In both toric and tropical contexts, I combined theoretical reasoning with extensive computational experiments. Using tools such as Python and Macaulay2, I implemented algorithms to test projective normality, compute syzygies, and verify covering properties of lattice polytopes. Looking ahead, I am eager to explore areas where mathematics interacts with computer science in the age of AI.

In conclusion, my academic path has been guided by curiosity about the structures underlying algebraic geometry and by persistence in tackling challenging problems. In the future, I hope to further broaden my understanding of the field, deepen my ability to conduct independent research, and contribute to advancing the frontiers of algebraic geometry through original and rigorous work.